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Design of a Quantum Self-Attention Neural Network on Quantum Circuits

Jin Zheng[1], Qing Gao[1*] and Zibo Miao[2]

Abstract— **This paper proposes a quantum self-attention neural network (QSAN) model that can be deployed on quantum circuits, providing a novel avenue to processing text classification tasks in natural language processing (NLP). The QSAN framework is established by integrating four basic blocks: the data preprocessing block, the quantum encoding block, the model design block, and the network optimization block. Simulation results demonstrate remarkable convergence and accuracy on various text classification datasets. In particular, the proposed QSAN surpasses the existing state-of-the-art quantum NLP (QNLP) model in terms of test accuracy.**

I. INTRODUCTION

Recent breakthroughs in quantum technologies have made noisy intermediate-scale quantum (NISQ) devices with stable computing performance accessible and established the implementation of quantum machine learning (QML) algorithms as a crucial research topic in industry and academia [1]. As a fundamental tool of QML, the quantum neural network (QNN) capitalizes on both the power of quantum computation and the effectiveness of machine learning, aiming to employ parameterized quantum circuits (PQCs) to address machine learning problems [2]. Recently, various categories of QNNs have been developed and effectively implemented in fields such as computer vision [3], [4] and reinforcement learning [5]. However, the research of QNNs in the NISQ era for AI applications remains in an embryonic stage, indicating ample room for further exploration and development.

As a research frontier in AI, natural language processing (NLP) is a vital subfield aiming to equip machines with the ability to comprehend human language. Despite advancements in tasks like machine translation, text classification, and speech recognition, the high complexity and flexibility of human language still pose challenges for NLP implementation [6]. Consequently, a natural question arises: How can quantum computing enhance a machine's performance in NLP? More specifically, how can we develop a quantum NLP (QNLP) model to tackle QNLP tasks? Due to the immense efforts of numerous researchers, several important works have addressed this issue, leading to the proposal of multiple QNLP models [7]. However, based on syntactic analysis, most of these QNLP proposals employ PQCs for sentences with different syntactical structures, resulting in limited flexibility and scalability. To address these drawbacks in current QNLP models, Li et al. proposed a Gaussian projection-based quantum self-attention neural network model for text classification [8]. Despite its significance, this work is limited by the underutilization of PQCs, as the computation of attention scores on classical computers is required to obtain the final output. Therefore, QNLP still faces several challenges: (i) The currently available PQC platforms provide only a limited number of qubits, basic linear quantum gates, and preliminary operations. Using these fundamental elements to construct a QNLP model for complex semantic analysis remains difficult. (ii) Designing a QNLP model with strong generalization and scalability to process sentences with diverse semantic structures remains an open challenge. To the best of our knowledge, only a few existing studies tackle these critical problems, thereby motivating this research.

In this paper, we propose a quantum self-attention neural network (QSAN) model designed to be deployed on PQCs, addressing the aforementioned challenges. The proposed technical framework is composed of four blocks: the data preprocessing block, the quantum encoding block, the model design block, and the network optimization block. Specifically, a Word2Vec model is employed to process classical text data into fixed-dimensional word vectors. Then, the amplitude encoding method is utilized to convert these word vectors into quantum states. Subsequently, a QSAN model based on PQCs is established to carry out the training task using the obtained quantum states. Finally, a quantum stochastic gradient descent (QSGD) algorithm is used to optimize the model parameters. Simulation results on various text classification datasets demonstrate the accuracy and generalization of the QSAN model. The main contributions of our research include the following:

- The proposed QSAN model, constructed using shallow quantum circuits, exhibits good convergence and accuracy on various datasets. It is shown that the QSAN can learn semantics between words and effectively process text classification tasks; and
- QSAN surpasses the existing state-of-the-art QNLP models in test accuracy, demonstrating scalability and the ability to handle various NLP tasks with minimal parameters. It is shown that the proposed QSAN has potential quantum advantages.

The structure of this paper is as follows: Section II covers

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the preliminaries. Section III explains the comprehensive design of the QSAN model. Section IV discusses the simulation settings and results, while Section V summarizes the paper.

II. Preliminaries

This section introduces the essential concepts of quantum computing and quantum machine learning models.

A. Quantum Computing

In quantum computing, the qubit serves as the basic element for information storage and processing, akin to the role of a bit in classical computing. A qubit lives in a 2-dimensional Hilbert space, established by two orthogonal bases quantum states, denoted as $|0\rangle$ and $|1\rangle$. Due to the principle of quantum superposition, a qubit can exist in any superposition state of these two basis states, represented as:

$$|\psi\rangle = a_0|0\rangle + a_1|1\rangle, \tag{1}$$

where the coefficients $a_0, a_1 \in \mathbb{C}$ satisfy $|a_0|^2 + |a_1|^2 = 1$.

Multiple qubits constitute an integrated quantum system, with the quantum state space of this system being a tensor product of each qubit state space, articulated as:

$$|\Psi\rangle = |\psi_0\rangle \otimes \ldots \otimes |\psi_{n-1}\rangle. \tag{2}$$

Analogously, the quantum state space for an n-qubit system comprises a set of states $|00\ldots0\rangle, \ldots, |11\ldots1\rangle$, and the quantum state $|\Psi\rangle$ can be represented as a superposition state of these bases:

$$|\Psi\rangle = \sum_{i=0}^{2^n-1} a_i|i_b\rangle, \tag{3}$$

where $|i_b\rangle$ denotes the quantum state associated with the binary representation of the integer i, and the coefficients a_i satisfy $\sum_{i=0}^{2^n-1} |a_i|^2 = 1$.

Just as classical logic gates operate in classical computing, quantum gates are used to manipulate the quantum states of qubits in quantum computing. In Hilbert space, a quantum gate is associated with a unitary operator U that meets the condition $UU^\dagger = I$. The application of a quantum gate to a superposition state behaves like matrix-vector multiplication:

$$U|\Psi\rangle = U\sum_{i=0}^{2^n-1} a_i|i_b\rangle = \sum_{i=0}^{2^n-1} b_i|i_b\rangle, \tag{4}$$

where the coefficients b_i satisfy $\sum_{i=0}^{2^n-1} |b_i|^2 = 1$. In practice, single-qubit rotation gates and CNOT gates are commonly employed in PQCs. The notations and representations of these gates can be found in Table I.

Quantum measurement is essential for extracting classical data from quantum states. The operation of projection measurement is commonly employed. Given a quantum observable M, a Hermitian operator, which adheres to the following spectral decomposition:

$$M = \sum_i \lambda_i P_{\lambda_i}, \tag{5}$$

TABLE I: The Quantum Logic Gates

Notations	Matrix representation	Graph representation
$R_x(\theta)$	$R_x(\theta) = \begin{bmatrix} \cos\frac{\theta}{2} & -i\sin\frac{\theta}{2} \\ -i\sin\frac{\theta}{2} & \cos\frac{\theta}{2} \end{bmatrix}$	$R_x(\theta)$
$R_y(\theta)$	$R_y(\theta) = \begin{bmatrix} \cos\frac{\theta}{2} & -\sin\frac{\theta}{2} \\ \sin\frac{\theta}{2} & \cos\frac{\theta}{2} \end{bmatrix}$	$R_y(\theta)$
$R_z(\theta)$	$R_z(\theta) = \begin{bmatrix} e^{-i\frac{\theta}{2}} & 0 \\ 0 & e^{i\frac{\theta}{2}} \end{bmatrix}$	$R_z(\theta)$
CNOT	$\mathrm{CNOT} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix}$	

here, P_{λ_i} signifies the projector onto the eigenspace of M that corresponds to the eigenvalue λ_i. The possible measurement outcomes are associated with the eigenvalues, λ_i, of the observable. When measuring the state $|\psi\rangle$, the probability of obtaining the outcome λ_i is calculated as:

$$p(\lambda_i) = \langle\psi\,|P_{\lambda_i}|\,\psi\rangle\,. \tag{6}$$

In particular, the expectation value of the observable M can be indicated as $\langle M\rangle = \langle\psi|M|\psi\rangle$.

B. Quantum Machine Learning Models

In this paper, the QML models discussed are QNNs that can be implemented using PQCs, as illustrated in Fig. 1.

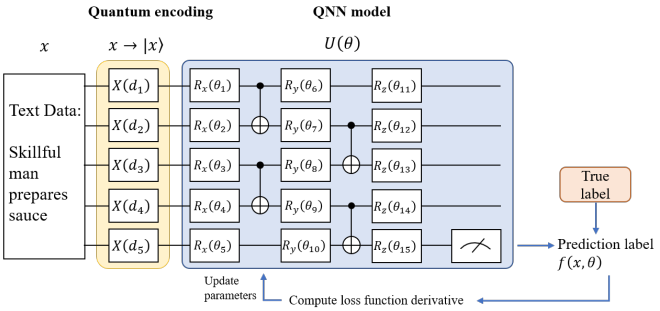


Fig. 1: The QML model on PQCs.

First, the quantum encoding block, marked in yellow, encodes the preprocessed classical input data x into its corresponding quantum state $|x\rangle$. Second, utilizing the blue-marked PQCs, a QNN architecture is built, which applies a unitary transformation $U(\theta)$ onto the input quantum state $|x\rangle$, wherein θ represents the tunable parameter vector. Following this transformation, quantum measurements are executed, yielding an output such as a prediction label denoted by $f(x,\theta)$. Third, this QNN output is utilized to calculate a loss function by comparing the prediction label with the true label. Finally, the resulting loss function is employed to adjust the parameters, and the updated parameters are fed back into $U(\theta)$ for the next iteration. This continuous iteration between the quantum and classical hardware is repeated multiple times to train the QNNs effectively.

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III. Quantum Self-Attention Neural Network Framework

This section presents the QSAN framework, which comprises four blocks: data preprocessing, quantum encoding, QSAN model, and network optimization, as shown in Fig. 2.

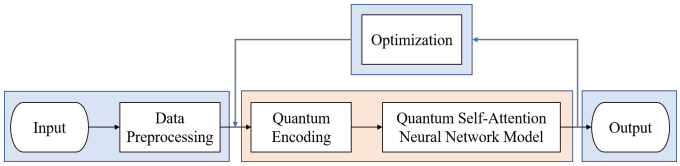


Fig. 2: The QSAN framework. Blue block: classical computer operations, orange block: quantum device operations.

A. Data Preprocessing

In this paper, QSAN is utilized to process text data, which inherently consists of natural language. To enable processing by a classical computer, the text data must be represented as a multidimensional vector. Two prevalent text representation methods are the one-hot encoding and the distributed representation model. In the one-hot representation approach, each word is assigned a vector with a length equivalent to the vocabulary size, where the element at the word index is one and all others are zero. Although this method is straightforward, its resource demands escalate substantially with the growth of the vocabulary size.

As another important text representation approach, the distributed representation method depicts a word through various textual properties, such as semantic meaning, tense, and similarity. Word2Vec is the most widely used algorithm in the distributed representation approach, as it efficiently produces word vectors by training on an extensive unsupervised text corpus. This algorithm includes two primary models: the Skip-gram and continuous bag of words (CBOW). The CBOW aims to anticipate the target word by considering its neighboring context, whereas the Skip-gram model performs the reverse operation. This paper employs the CBOW model to generate word vectors, contributing to a more robust and coherent understanding of the text.

Fig. 3 illustrates the four steps of the CBOW model for generating word vectors:

Step 1: Inputs are one-hot representations of words in sentences with C words. For word w_{1k}, the input is $[x_{11}, \ldots, x_{1k}, \ldots, x_{1V}]$, where $x_{1k} = 1$, others are 0, and V is the vocabulary size.

Step 2: A weight matrix $W_{V\times N}$ between input and hidden layers is initialized, with N as the hidden layer dimension.

Step 3: A weight matrix $W'_{N\times V}$ between hidden and output layers is initialized.

Step 4: A softmax function is deployed at the final layer to deduce the probability distribution of the target word based on its surrounding context words.

The goal of training the CBOW model is to adjust the weights of the two matrices to maximize the likelihood of the target word given its context words. The optimized weight $W_{V\times N}$ represents the word vector.

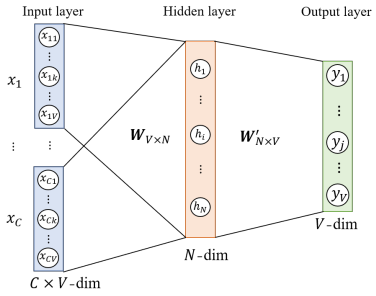


Fig. 3: The CBOW model.

B. Quantum Encoding

To process classical data using quantum computing platforms, a quantum encoding block is essential for converting the classical input data x into the quantum format $|x\rangle$, as shown in Fig. 2. The two main quantum encoding techniques are basis encoding and amplitude encoding. Compared to the amplitude encoding approach, the basis encoding method is simpler to implement, but it requires a significant increase in the number of qubits used when dealing with high-precision floating-point numbers. Therefore, this paper adopts the amplitude encoding approach.

The amplitude encoding approach transforms classical data into the probability amplitudes of a quantum state. With this approach, a system containing n qubits has 2^n amplitudes capable of storing 2^n real-number values. Given a 2^n dimensional real-number vector $x = [x_1, \ldots, x_{2^n}]^T$ with elements x_i, the corresponding quantum state is expressed as:

$$|x\rangle = \frac{1}{C}\sum_{i=1}^{2^n} x_i |i_b\rangle, \tag{7}$$

where $C = \sqrt{\sum_{i=1}^{2^n} |x_i|^2}$ serves as the normalization constant. For example, $x = [-1.0, 0.0, 2.5, 1.0]^T$ is represented by the 2-qubit quantum state $|x\rangle = \frac{1}{\sqrt{6.25}}(-|00\rangle + 2.5|10\rangle + |11\rangle)$. It is noteworthy that the storage capability of a quantum system escalates exponentially with the count of qubits in the amplitude encoding approach.

C. Quantum Self-Attention Neural Network Model

a) Classical Self-Attention Neural Network Model: The classical self-attention neural network (CSAN) model utilizes the self-attention mechanism to achieve powerful representational learning and efficient parallel computation. Through the self-attention mechanism, the model can directly discern associations between arbitrary pairs of elements in the input sequence, thereby enabling effective apprehension of long-range dependencies and enhancing the precision of predictions and classifications.

The self-attention mechanism, a crucial component of CSAN models, employs the attention function to map a query and associated key-value pairs into an output vector.

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This output is generated through a weighted summation of value vectors, with weights assigned based on a compatibility function between the query and corresponding keys.

Formally, given an input sequence with n words, $\{x_i \in \mathbb{R}^d\}_{i=1}^n$, each word x_i is transformed using three weight matrices, W^q, W^k, $W^v \in \mathbb{R}^{d\times d}$, resulting in three components: the query $W^q x_i$, the key $W^k x_i$, and the value $W^v x_i$, respectively. A compatibility score e_{ij} is computed using a function like the scaled dot product, comparing $W^q x_i$ and $W^k x_j$:

$$e_{ij} = \frac{(W^q x_i)\left(W^k x_j\right)^T}{\sqrt{d}}. \tag{8}$$

Subsequently, the attention weight α_{ij} is calculated to represent the relevance between different words in the sequence:

$$\alpha_{ij} = \frac{\exp(e_{ij})}{\sum_{l=1}^{n}\exp(e_{il})}. \tag{9}$$

Finally, the output representation, $\{y_i \in \mathbb{R}^d\}_{i=1}^n$, is constructed as a weighted sum of the transformed values:

$$y_i = \sum_{j=1}^{n} \alpha_{ij}\left(W^v x_j\right). \tag{10}$$

The self-attention mechanism above is utilized to construct a self-attention layer. By integrating a fully connected layer, a comprehensive CSAN architecture is established.

b) Quantum Self-Attention Model: Inspired by the CSAN model, a universal QSAN model has been developed by designing PQCs that consist of a quantum self-attention layer and a quantum fully connected layer, as illustrated in Fig. 4.

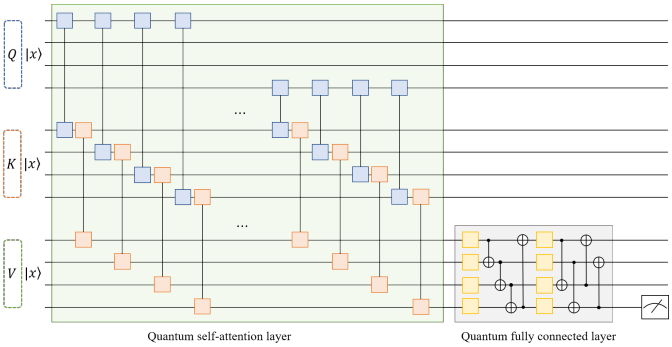


Fig. 4: The quantum self-attention neural network model.

- Quantum self-attention layer

In Fig. 4, the quantum self-attention layer is constructed using a strongly entangled quantum circuit block and a universal 2-qubit gated block.

(i) The strongly entangled quantum circuit block

In the quantum self-attention layer, strongly entangled quantum circuits generate highly entangled quantum states to effectively capture the correlation between the query $Q\left(W^q x\right)$ and the key $K\left(W^k x\right)$. These circuits are constructed by applying single-qubit gates to each qubit individually and subsequently interconnecting them with controlled single-qubit gates based on their control range [9]. Fig. 5 illustrates two strongly entangled quantum circuit blocks, B_1 and B_2, where G represents single-qubit gates, with control ranges of 1 and 2 for the controlled single-qubit gates, respectively.

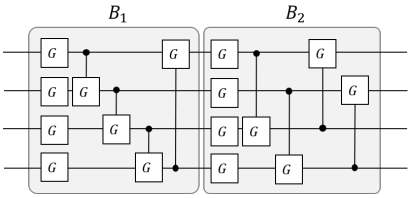


Fig. 5: The universal strongly entangled quantum circuits.

The strongly entangled quantum circuit block is crucial in the quantum self-attention layer. Specifically, this block captures the relationship between Q and K while representing the weights W^q and W^K acting on the state $|x\rangle$. To achieve this objective, a parameterized universal single-qubit gate, denoted as G_u, is introduced, which can be decomposed into a combination of R_z and R_y rotation gates. To maintain strong entanglement capabilities and reduce the model's circuit depth, the CNOT gate is selected as the controlled single-qubit gate. Consequently, the strongly entangled block within the quantum self-attention layer comprises single-qubit rotation gates R_y, R_z, and the CNOT gate, as depicted in Fig. 6.

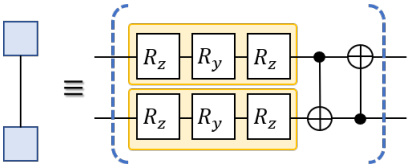


Fig. 6: The strongly entangled quantum citcuits block.

(ii) The universal 2-qubit gate block

The entanglement quantum circuit block establishes a correlation between Q and K. Subsequently, the result is assigned to $V(W^v x)$ to construct a new quantum state representation, and this transformation is realized through the implementation of a universal 2-qubit gate. However, due to the limited availability of basic quantum gates in real quantum computing hardware, the universal 2-qubit gate is decomposed into a series of fundamental quantum gates, specifically from the set $\{CNOT, R_y, R_z\}$, as demonstrated in Fig. 7.

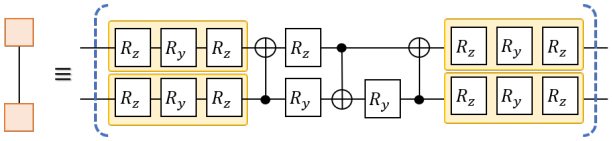


Fig. 7: The universal 2-qubit gate.

This architecture can generate new words by incorporating information from other words in the sentence, effectively

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capturing their dependency relationships and coherently implementing the self-attention mechanism.

- Quantum fully connected layer

Similar to the fully connected layer in the CSAN model, a quantum fully connected layer is implemented in the quantum circuit representing V after the quantum self-attention layer for additional feature mapping.

Fig. 4 presents the designed quantum fully connected layer, which consists of two strongly entangled quantum circuit blocks operating on numerous qubits. The utilization of two blocks is attributed to the increased qubit count compared to the quantum self-attention layer.

D. Network Optimization

Upon obtaining the classical measurement results, one can execute optimization algorithms and complete the model training process. Analogous to the gradient-based optimization methods used in CSAN, QSGD is employed for optimizing the QSAN model.

Before proceeding, the notation used in this section is defined as follows: Let $D^{\text{train}} = \left\{x_i^{\text{train}}, y_i^{\text{train}}\right\}_i^S$ and $D^{\text{test}} = \left\{x_j^{\text{test}}, y_j^{\text{test}}\right\}_j^Q$ represent the training and test sets, respectively. Here, $x_i^{\text{train}}, x_j^{\text{test}} \in \mathbb{R}^N, y_i^{\text{train}}, y_j^{\text{test}} \in \{-1, 1\}$, and S and Q indicates the respective set sizes. The loss function is given by $\ell = \frac{1}{S}\sum_{i=1}^{S}\left(y_i - \ y_i^{\text{train}}\right)^2$, where y_i is the QSAN model's classical output.

In training QSAN, it is challenging to calculate gradients using backpropagation, commonly used in classical neural networks, due to the inability to store and utilize intermediate quantum operation results on hardware. To address this issue, the parameter-shift rule can directly compute gradients on quantum hardware. A detailed implementation can be found in [10]. Once the gradient is obtained, the QSGD algorithm can update the parameters and optimize the QSAN model.

Algorithm 1 QSGD Algorithm

Require: Quantum input states $\left\{|x_i^{train}\rangle\right\}_{i=1}^S$ for the dataset $\left\{x_i^{\text{train}}, y_i^{\text{train}}\right\}_{i=1}^S$, the QSAN model with parameters θ, the loss function ℓ, the maximum number of iterations T, and the threshold δ.

Ensure: The optimized parameters θ^*.

1: Set t = 0 and randomly initialize each parameter in $\theta^{(t)}$ within the interval $[0, 2\pi]$. Set the learning rate η.
2: **while** $\left|\left|\ell\left(\theta^{(t)}\right)\right|\right| \geq \delta$ and $t \leq T$ **do**
3: Randomly select a training sample $\left\{|x_i\rangle^{\text{train}}, y_i^{\text{train}}\right\}$.
4: Compute the gradient $\nabla\ell\left(\theta^{(t)}\right)$ utilizing the parameter-shift rule.
5: Update the parameters $\theta^{(t+1)} = \theta^{(t)} - \eta \cdot \nabla\ell\left(\theta^{(t)}\right)$.
6: $t \leftarrow t + 1$.
7: **end while**
8: Return the optimized parameters θ^*.

IV. Simulation

In this section, the proposed QSAN model is employed for sentence classification tasks on two distinct NLP datasets, with data preprocessing and performance analysis illustrating its efficacy and generalizability.

A. Data Preprocessing

In the simulation, two synthetic datasets, MC and RP, were utilized, both derived from [11]. The MC dataset was generated using a simple context-free grammar and consists of an equal proportion of food-related and IT-related sentences. The RP dataset was obtained from the RELPRON dataset and is designed to predict the presence of subject-based or object-based relative clauses within noun phrases. Specifically, the MC dataset contains 17 unique words in 130 sentences (70 for training, 30 for development, and 30 for testing), with each sentence having 3 or 4 words. The RP dataset contains 115 unique words in 105 sentences (74 for training and 31 for testing), with each sentence containing 4 words.

In particular, two illustrative examples are presented for the MC and RP datasets, respectively.

1) "person prepares tasty meal."
   "woman runs program."
2) "organization that soldier serve."
   "scientist that visit island."

The data preprocessing discussed involves two steps:

Step 1: Converting MC and RP data into fixed-length word vectors. The CBOW model described in Section III. A is employed for encoding words in a sentence into fixed-length vectors. In the simulation, both sentence length and vector dimension are set to 4. For sentences in the MC dataset with a size of 3, padding is applied by adding a 4-dimensional zero vector. As a result, all sentences are transformed into 16-dimensional vectors.

Step 2: Encoding word vectors into quantum states. Normalized word vectors from Step 1 are encoded into quantum states using the amplitude encoding method, enabling the representation of a 16-dimensional word vector with 4 qubits.

B. Simulation Settings

Table II summarizes the simulation settings used in this paper.

TABLE II: Simulation Settings

Datasets	MC and RP
Programming platform	PennyLane
Number of qubits used	12
Optimization algorithm	QSGD
Learning rate	0.1
Number of training iterations	1000

C. Comparative Models

Currently, two works related to our research are compared to the QSAN proposed in the simulation.

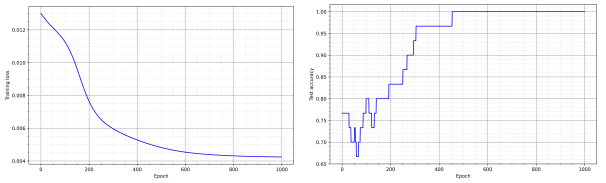
1) **DisCoCat model:** This is the first QNLP model that integrates quantum circuits with NLP, achieving simple sentence classification and laying the foundation for subsequent QNLP research.
2) **QSANN model:** This method adopts Gaussian projected quantum self-attention to construct a quantum

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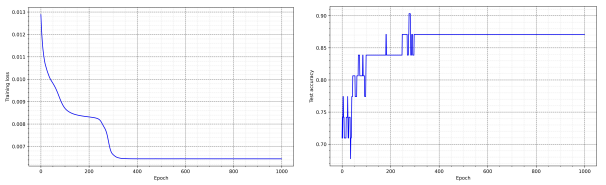
self-attention neural network. It has the potential advantage of uncovering hidden correlations between words that are difficult to discover using classical methods.

D. Simulation Results

In the simulation, training loss and test accuracy are evaluation metrics for the sentence classification task. The QSAN model was tested on both the MC and RP datasets, with the simulation results displayed in Fig. 8.



(a) The training loss of MC. (b) The test accuracy of MC.



(c)The training loss of RP. (d) The test accuracy of RP.

Fig. 8: The simulation results.

One can observe from Fig. 8(a) and Fig. 8(c) that the QSAN model converges on both datasets after 1000 epochs, indicating its effectiveness for the sentence classification task. One can also observe from Fig. 8(b) and Fig. 8(d) that the QSAN model achieves stable and high test accuracy. In particular, the QSAN model quickly achieves 100% test accuracy on the MC dataset. However, the test accuracy on the RP dataset is slightly lower at 87.1% compared to the MC dataset. This is mainly due to the characteristics of the RP dataset, which has a large vocabulary but a small data size, leading to a significant bias between the training and test sets, i.e., many words only appear in the test set and not in the training set.

TABLE III: Simulation Results

Method	MC		RP	
	#Parameters[1]	Test accuracy	#Parameters[1]	Test accuracy
DisCoCat	40	79.8%	168	72.3%
QSANN	25	100%	109	67.74%
QSAN	45	**100%**	45	**87.10%**

[1] #Parameters means the number of parameters.

The comparative results of QSAN, DisCoCat, and QSANN models are summarized in Table III. From the test accuracy perspective, QSAN effortlessly achieves 100% test accuracy in the MC task, which is the same as QSANN and 20.2% higher than DisCoCat. In the RP task, QSAN obtains the highest test accuracy. These results demonstrate that QSAN possesses exceptional classification performance for sentence classification tasks. From the number of parameters perspective, the QSAN model contains 45 parameters, with 21 in the quantum self-attention layer and 24 in the quantum fully connected layer. One can observe from Table III that QSAN uses the same network structure of different datasets with the same dimensional word vector. Due to the simplistic features of the MC task, QSAN requires a larger number of parameters than DisCoCat and QSANN. However, its advantages are significant when dealing with complex data features like those in the RP task. Thus, QSAN can handle multiple tasks with fewer parameters and exhibits strong generalization performance.

V. Conclusion

In this paper, a novel QSAN model has been proposed for addressing text classification tasks in NLP. The model has been designed with four fundamental blocks: the data preprocessing block that processes classical text data into fixed-dimensional word vectors; the quantum encoding block that encodes word vectors into quantum states; the model design block that captures semantic relationships between words; and the network optimization block that exhibits outstanding optimization capabilities. Simulations have been implemented to showcase the excellent performance of the QSAN model. Further research will focus on constructing multi-head QSANs to extract more complex semantic information between words.

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